

## POINT OF AYRE LH, Isle of Man

WMO: 032080

Lat: 54.417N

Lon: 4.367W

Elev: 9

StdP: 101.22

Time Zone: 0.00 (EUW)

Period: 86-02

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 0.6        | 1.5 | -4.4                          | 2.6 | 2.1  | -3.2 | 2.9 | 2.6  | 21.4                  | 6.4  | 19.7 | 7.4  | 6.1                   | 120  | 0.794 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 4.9                    | 20.3            | 16.2 | 19.2 | 15.6 | 18.4 | 15.1 | 17.1                | 19.1 | 16.4 | 18.3 | 15.7 | 17.6 | 5.8                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|----------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)            |
| 16.3                            | 11.6 | 18.1 | 15.5 | 11.0 | 17.4 | 14.8 | 10.5 | 16.8 | 48.1          | 19.2 | 46.0 | 18.2 | 44.1 | 17.8 | 19.8           |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 17.1          | 14.9 | 13.6 | -1.7                       | 23.6 | 1.6                | 1.3 | -2.9                                               | 24.5 | -3.8       | 25.2 | -4.7       | 25.9 | -5.9       | 26.9 |      |     |
| (5)               |      |      | DB                         | -2.5 | 18.9               | 2.1 | 0.8                                                | -4.0 | 19.5       | -5.2 | 20.0       | -6.4 | 20.5       | -7.9 | 21.1 | (5) |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |      |     |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.2   | 5.9   | 5.9   | 6.8   | 8.0   | 11.0  | 13.2  | 15.2  | 15.4  | 13.9  | 11.5  | 8.7   | 6.8  |
|                                                                                  | DBStd       | 4.02   | 2.36  | 2.42  | 2.19  | 1.91  | 2.02  | 1.94  | 1.51  | 1.69  | 1.88  | 2.24  | 2.06  | 2.30 |
|                                                                                  | HDD10.0     | 586    | 127   | 117   | 102   | 64    | 13    | 1     | 0     | 0     | 1     | 11    | 49    | 101  |
|                                                                                  | HDD18.3     | 2970   | 384   | 349   | 358   | 310   | 228   | 155   | 99    | 92    | 134   | 212   | 290   | 357  |
|                                                                                  | CDD10.0     | 660    | 1     | 1     | 2     | 4     | 43    | 96    | 159   | 168   | 116   | 57    | 9     | 3    |
|                                                                                  | CDD18.3     | 2      | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 1     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 2      | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 1     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 7.1    | 8.6   | 8.7   | 7.8   | 6.4   | 5.9   | 5.8   | 5.7   | 5.7   | 6.4   | 7.7   | 8.1   | 8.4  |
| Precipitation                                                                    | PrecAvg     | 838    | 84    | 58    | 62    | 54    | 49    | 49    | 50    | 70    | 76    | 99    | 96    | 89   |
|                                                                                  | PrecMax     | 1004   | 185   | 129   | 120   | 107   | 136   | 106   | 129   | 140   | 136   | 184   | 163   | 164  |
|                                                                                  | PrecMin     | 669    | 16    | 1     | 14    | 7     | 10    | 15    | 13    | 8     | 7     | 16    | 28    | 13   |
|                                                                                  | PrecStd     | 92     | 36    | 33    | 27    | 27    | 30    | 23    | 30    | 30    | 34    | 46    | 32    | 35   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 12.3  | 12.7  | 13.0  | 14.7  | 18.3  | 21.4  | 21.9  | 21.7  | 20.2  | 17.5  | 14.5  | 13.5 |
|                                                                                  |             | MCWB   | 10.6  | 10.5  | 10.6  | 11.8  | 14.3  | 16.3  | 16.8  | 17.5  | 16.3  | 13.9  | 13.0  | 12.3 |
|                                                                                  | 2%          | DB     | 11.1  | 11.5  | 11.8  | 13.3  | 16.6  | 19.0  | 20.2  | 20.1  | 18.6  | 16.2  | 13.3  | 12.0 |
|                                                                                  |             | MCWB   | 9.7   | 9.7   | 9.9   | 10.7  | 13.2  | 14.7  | 16.2  | 16.4  | 14.9  | 13.7  | 12.0  | 11.0 |
|                                                                                  | 5%          | DB     | 10.3  | 10.6  | 10.9  | 12.1  | 15.3  | 17.5  | 19.1  | 19.1  | 17.6  | 15.5  | 12.6  | 11.1 |
|                                                                                  |             | MCWB   | 9.1   | 9.1   | 9.3   | 9.8   | 12.3  | 14.1  | 15.5  | 16.0  | 14.2  | 13.3  | 11.3  | 10.1 |
|                                                                                  | 10%         | DB     | 9.5   | 9.6   | 10.0  | 11.1  | 14.3  | 16.5  | 18.1  | 18.3  | 16.7  | 14.7  | 11.8  | 10.2 |
|                                                                                  |             | MCWB   | 8.4   | 8.2   | 8.6   | 9.2   | 11.6  | 13.6  | 15.0  | 15.5  | 13.9  | 12.8  | 10.7  | 9.3  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 11.0  | 10.6  | 11.1  | 12.4  | 14.9  | 16.8  | 18.0  | 18.4  | 17.2  | 15.3  | 13.3  | 12.5 |
|                                                                                  |             | MCDB   | 11.7  | 11.7  | 12.5  | 14.6  | 17.9  | 20.4  | 20.7  | 20.6  | 18.7  | 16.2  | 14.0  | 13.3 |
|                                                                                  | 2%          | WB     | 10.1  | 9.9   | 10.3  | 11.2  | 13.7  | 15.4  | 16.7  | 17.4  | 16.1  | 14.6  | 12.3  | 11.2 |
|                                                                                  |             | MCDB   | 11.0  | 11.0  | 11.4  | 12.8  | 16.0  | 18.1  | 19.3  | 19.0  | 17.6  | 15.6  | 13.0  | 11.8 |
|                                                                                  | 5%          | WB     | 9.3   | 9.1   | 9.6   | 10.4  | 12.7  | 14.7  | 16.0  | 16.6  | 15.2  | 13.8  | 11.6  | 10.4 |
|                                                                                  |             | MCDB   | 10.2  | 10.1  | 10.6  | 11.8  | 14.7  | 17.0  | 18.3  | 18.2  | 16.6  | 14.9  | 12.3  | 11.0 |
|                                                                                  | 10%         | WB     | 8.6   | 8.2   | 8.8   | 9.6   | 12.0  | 14.0  | 15.4  | 16.0  | 14.5  | 13.1  | 10.8  | 9.4  |
|                                                                                  |             | MCDB   | 9.5   | 9.2   | 9.8   | 10.9  | 13.9  | 16.0  | 17.4  | 17.7  | 16.1  | 14.3  | 11.7  | 10.1 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 3.2   | 3.6   | 3.9   | 4.7   | 5.4   | 5.2   | 5.1   | 4.9   | 4.5   | 3.5   | 3.1   | 2.9  |
|                                                                                  |             | MCDBR  | 4.3   | 4.3   | 4.7   | 6.2   | 6.9   | 7.0   | 6.9   | 6.0   | 5.3   | 4.4   | 3.9   | 3.6  |
|                                                                                  |             | MCWBR  | 4.3   | 4.1   | 3.7   | 4.3   | 4.1   | 4.1   | 3.8   | 3.5   | 3.5   | 3.4   | 4.0   | 3.8  |
|                                                                                  | 5% WB       | MCDBR  | 4.1   | 4.4   | 4.5   | 5.7   | 6.5   | 6.4   | 6.2   | 5.5   | 4.9   | 3.8   | 3.9   | 3.6  |
|                                                                                  |             | MCWBR  | 4.2   | 4.4   | 3.9   | 4.4   | 4.3   | 4.2   | 3.9   | 3.7   | 4.0   | 3.5   | 4.1   | 3.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.333  | 0.344 | 0.369 | 0.387 | 0.391 | 0.385 | 0.387 | 0.373 | 0.365 | 0.350 | 0.327 | 0.319 |      |
|                                                                                  | taud        | 2.280  | 2.363 | 2.336 | 2.320 | 2.367 | 2.412 | 2.410 | 2.463 | 2.482 | 2.452 | 2.358 | 2.259 |      |
|                                                                                  | Ebn at Noon | 597    | 739   | 802   | 838   | 855   | 863   | 854   | 846   | 805   | 726   | 607   | 525   |      |
|                                                                                  | Edh at Noon | 60     | 77    | 97    | 112   | 113   | 109   | 108   | 97    | 84    | 70    | 55    | 50    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.56   | 1.25  | 2.44  | 3.93  | 5.04  | 5.11  | 4.88  | 4.03  | 2.80  | 1.53  | 0.72  | 0.42  |      |
|                                                                                  | RadStd      | 0.06   | 0.12  | 0.19  | 0.30  | 0.41  | 0.48  | 0.41  | 0.29  | 0.19  | 0.16  | 0.07  | 0.04  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (62 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page